An Analysis Plan and Uncertainty Quantification Framework for the sPHENIX π^0 Nuclear Modification Factor R_{CP}

with Integrated AI-Agent Workflow Design

Parker Lewis
Ohio University, Department of Physics & Astronomy
Advisor: Prof. Justin Frantz
sPHENIX and STAR Collaborations

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Abstract

This document is a thesis-grade methodological framework for the sPHENIX neutral pion central-to-peripheral nuclear modification factor $R_{\mathrm{CP}}(p_T, \mathrm{cent})$. It provides (i) a rigorous mathematical and statistical treatment, (ii) a fully cataloged systematic uncertainty plan benchmarked against PHENIX, STAR, and ALICE precedents, (iii) closure and validation prescriptions, and (iv) a deliberate division of labor between physicist judgment and Large-Language-Model (LLM) agentic assistance, anchored to recent HEP automation work [28]. The intent is that the document serve simultaneously as an internal analysis note skeleton, a thesis chapter outline, and a project-management instrument.

Contents

1 Executive Summary and Scope

The sPHENIX detector at RHIC has, since the 2023 commissioning run, demonstrated π^0 reconstruction in the EMCal in the standard di-photon decay channel [22]. With the Run-25 Au+Au dataset, sPHENIX is positioned to extend the high-precision RHIC π^0 suppression program first established by PHENIX [1, 2, 3]. The flagship single-particle hard-probe observable is the nuclear modification factor:

$$R_{\mathrm{AA}}(p_T) \;=\; \frac{(1/N_{\mathrm{evt}}^{AA})\, dN_{\pi^0}^{AA}/dp_T}{\langle T_{AA}\rangle\, d\sigma_{\pi^0}^{pp}/dp_T},$$

which compares heavy-ion yields to a binary-collision-scaled *pp* baseline. In its absence (or while a *pp* analysis is being finalized), the central-to-peripheral ratio

$$R_{\mathrm{CP}}(p_T) \;=\; \frac{\langle N_{\mathrm{coll}}^{\mathrm{peri}}\rangle}{\langle N_{\mathrm{coll}}^{\mathrm{cent}}\rangle} \cdot \frac{(1/N_{\mathrm{evt}}^{\mathrm{cent}})\, dN_{\pi^0}^{\mathrm{cent}}/dp_T}{(1/N_{\mathrm{evt}}^{\mathrm{peri}})\, dN_{\pi^0}^{\mathrm{peri}}/dp_T}$$

provides a methodologically equivalent, internally referenced suppression measurement that benefits from cancellation of detector-level systematics and luminosity normalization. PHENIX, STAR, and ATLAS have all employed R_{CP} as a primary or cross-check observable [2, 15, 16].

This document is organized to address two intertwined deliverables that the analyzer requires:

(A) Physics methodology.
How to construct, propagate, and document the statistical and systematic uncertainty budget for the sPHENIX π^0 R_{CP} measurement. This covers detector calibration (Tower Slope Method), signal extraction (Gauss+polynomial / power-law backgrounds with iterative sideband-based χ^2 minimization), efficiency / acceptance corrections (`pi0Efficiency` module), centrality and N_{coll}, and ratio-specific issues (peripheral-bin biases [13, 14]).

(B) Agentic workflow integration.
How to embed Claude / LLM-agent assistance into the workflow as a complement to, never a replacement for, physics judgment. The structure draws on recent prototypes such as the LLM-supervisor + Snakemake pipeline of Gendreau-Distler *et al.* [28], the document-grounded code-generation work of CoLLM-style systems [29], and the IRIS-HEP *LLMs as Assistants* effort [30].

Validation & Warning

Why uncertainty quantification matters specifically for this observable. The ratio of $R_{\mathrm{CP}} \sim 0.2$ at intermediate p_T is interpreted as a measure of the path-length integrated parton energy loss and constrains $\hat{q}$, dN^g/dy, and the medium opacity. Because the leading-order interpretation depends on ~ 0.1 in the deviation from unity, an uncorrelated systematic error on R_{CP} of $\sim 5{-}10\%$ already substantially weakens model discrimination. Worse, several systematics (notably the EMCal energy scale, N_{coll} in the peripheral bin, and the centrality-trigger bias) are *partially correlated* between numerator and denominator. Treating them as fully cancelling (optimistic) or as fully uncorrelated (conservative) can shift the interpretation by factor-of-two in significance. The plan below treats correlations explicitly via a full covariance construction.

2 Mathematical Framework

2.1 Definitions and primary observable

Key Formula

Per-event, per-centrality π^0 invariant yield:

$$Y_c(p_T) \;\equiv\; \frac{1}{N^c_{\mathrm{evt}}}\,\frac{1}{2\pi p_T\,\Delta y\,\Delta p_T}\,\frac{N^c_{\pi^0}(p_T)}{\varepsilon^c_{\mathrm{rec}}(p_T)\,A^c(p_T)\,\mathrm{BR}_{\pi^0\to\gamma\gamma}\,\varepsilon^c_{\mathrm{trig}}(p_T)}, \tag{1}$$

where c labels a centrality class, $N^c_{\pi^0}$ is the extracted raw yield in (p_T, c), $\varepsilon_{\mathrm{rec}}$ is the reconstruction efficiency (clustering + identification), A is the geometric acceptance×dead-channel correction, $\varepsilon_{\mathrm{trig}}$ is the trigger efficiency, $\mathrm{BR} = 0.98823$, and Δy is the rapidity bin width.

R_{CP} in terms of yields:

$$\boxed{R_{\mathrm{CP}}(p_T) \;=\; \frac{\langle N_{\mathrm{coll}}\rangle_{\mathrm{peri}}}{\langle N_{\mathrm{coll}}\rangle_{\mathrm{cent}}}\,\frac{Y_{\mathrm{cent}}(p_T)}{Y_{\mathrm{peri}}(p_T)}\;.} \tag{2}$$

The peripheral reference (typically 60–80% in PHENIX [1], with 60–92% sometimes shown as a cross-check) is chosen so that the plasma is most weakly produced, i.e. $R_{\mathrm{CP}} \to 1$ in the limit of no medium. Choices of peripheral interval $\{[60, 80]\%, [50, 80]\%, [40, 80]\%\}$ should be carried as a systematic variation (Sec. 3).

2.2 Statistical uncertainty propagation

For each (p_T, c) the raw yield $N^c_{\pi^0}$ is extracted from a fit to the $m_{\gamma\gamma}$ distribution, typically as

$$F(m;\boldsymbol{\theta}) \;=\; \underbrace{A\,\exp\Big[-\tfrac{1}{2}\big(\tfrac{m-\mu}{\sigma}\big)^2\Big]}_{\text{signal (Gaussian or CB)}} \;+\; \underbrace{\sum_{k=0}^{n} c_k m^k}_{\text{combinatorial bg.}} \tag{3}$$

with the iterative sideband-based χ^2 minimization that the analyzer's framework already implements. Two distinct statistical contributions enter:

1. **Counting (Poisson) uncertainty** on the integrated signal counts: $\sigma^N_{\rm stat} = \sqrt{N_S + B}$ for an integration window of total $S + B$ counts in a $\pm n\sigma$ window, where S is the signal and B the background under the peak.

2. **Fit-parameter uncertainty**: when yield is taken as $N_S = A\sqrt{2\pi}\,\sigma/\Delta m$ from a binned χ^2 or extended likelihood fit, propagate the parameter covariance matrix $V_{\boldsymbol{\theta}}$:

$$\sigma^2_{N_S} = (\nabla_{\boldsymbol{\theta}} N_S)^T\, V_{\boldsymbol{\theta}}\, (\nabla_{\boldsymbol{\theta}} N_S)\,. \tag{4}$$

For the ratio in (2), with independent Poisson statistics in the two centrality bins (a good approximation since they are disjoint event subsamples),

$$\left(\frac{\sigma^{\rm stat}_{R_{\rm CP}}}{R_{\rm CP}}\right)^2 \;=\; \left(\frac{\sigma^{\rm cent}_{N_{\pi^0}}}{N^{\rm cent}_{\pi^0}}\right)^2 + \left(\frac{\sigma^{\rm peri}_{N_{\pi^0}}}{N^{\rm peri}_{\pi^0}}\right)^2 . \tag{5}$$

The peripheral bin will dominate the statistical error in the highest p_T bins, since the π^0 yield in $[60, 80]\%$ is suppressed by $\sim N^{\rm peri}_{\rm coll}/N^{\rm cent}_{\rm coll} \sim 0.04$ relative to central. Bootstrap resampling of the unbinned $m_{\gamma\gamma}$ candidates is the cleanest cross-check and should be implemented (Sec. 4).

2.3 Bin migration

Even at sPHENIX EMCal resolution ($\sigma_E/E \approx 16\%/\sqrt{E} \oplus 5\%$ [21]), bin-to-bin migration is non-negligible above $p_T \gtrsim 10$ GeV/c, where the steeply falling spectrum amplifies any energy-scale or resolution mis-modeling. Two equivalent treatments are acceptable: (i) a Bayesian / SVD unfolding of $N^c_{\pi^0}(p_T)$ before forming the ratio, with the response matrix from full embedded MC; or (ii) a forward-folding correction $C_{\rm shape}(p_T)$ that accounts for the convolution of the true spectrum with the measured resolution. Method (ii), as used historically by PHENIX [1], is recommended for sPHENIX given its smaller dynamic range relative to ALICE / ATLAS.

2.4 Systematic categorization (type A/B, additive/multiplicative)

Following Barlow's taxonomy [23, 25], distinguish:

- **Type A (point-to-point uncorrelated):** variations whose pull is independent in each p_T bin. Examples: yield-extraction polynomial-order variation (different bins fit independently), bin-by-bin background normalization. These add in quadrature *within* a centrality class; for the ratio they propagate as in Sec. 2.2.

- **Type B (correlated within p_T, i.e. tilt/shift):** energy scale, energy resolution, η/π^0 feed-down, conversion correction. These are best represented as a coherent shape variation that shifts *all* p_T bins together.
- **Type C (global, normalization-only):** N_{coll}, integrated luminosity (irrelevant for R_{CP} since both numerator and denominator are per-event yields), trigger absolute scale.

The PHENIX papers from PRL 101 / PRC 87 / PRC 88 show this categorization explicitly via separate boxes drawn around each data point versus a global error band [1, 2].

2.5 Covariance matrix construction

For a vector of N_b measured R_{CP} values across p_T bins, define

$$\mathrm{Cov}^{\mathrm{tot}}_{ij} \;=\; \mathrm{Cov}^{\mathrm{stat}}_{ij} \;+\; \sum_{s\in\mathcal{S}} \rho^{(s)}_{ij}\, \sigma^{(s)}_i \sigma^{(s)}_j, \tag{6}$$

where $\mathcal{S}$ ranges over independent systematic sources, $\sigma^{(s)}_i$ is the size of source s in bin i, and $\rho^{(s)}_{ij} \in \{0, +1\}$ encodes correlation:

- Type A: $\rho_{ij} = \delta_{ij}$;
- Type B: $\rho_{ij} = +1$ for all i, j in the same coherent direction (or signed if shape-dependent);
- Type C: $\rho_{ij} = +1$ across all bins (a single global scale).

This $\mathrm{Cov}^{\mathrm{tot}}$ is what should be released alongside the central values when sharing data with theorists, exactly as ALICE has done in its π^0 R_{AA} HEPData releases [12, 8].

2.6 Treatment of N_{coll} uncertainty in R_{CP} vs. R_{AA}

Validation & Warning

This is the central methodological subtlety of R_{CP}. The Glauber-model N_{coll} uncertainty is dominated by:

1. Inelastic NN cross section $\sigma^{NN}_{\mathrm{inel}}$ (± 1 mb at 200 GeV);
2. Nuclear density profile (Woods–Saxon parameters);
3. Fluctuations and the NBD ancestor model;
4. Most importantly for sPHENIX: the MBD/sEPD trigger efficiency in *pp* used to anchor the centrality [17].

For R_{AA}, the dominant N_{coll} uncertainty in 0–10% Au+Au is roughly $\pm 7\%$ for PHENIX. For R_{CP}, the relevant quantity is the *ratio* $R_{\mathrm{coll}} = \langle N^{\mathrm{cent}}_{\mathrm{coll}}\rangle / \langle N^{\mathrm{peri}}_{\mathrm{coll}}\rangle$, which for $[0, 10]\%/[60, 80]\%$ is ~ 23. The trigger-efficiency component [17] *does not cancel*: in fact it grows binomially as one moves to peripheral bins, where a 2–3% trigger inefficiency in *pp* translates into a $\gtrsim 20\%$ uncertainty on $\langle N^{\mathrm{peri}}_{\mathrm{coll}}\rangle$. This is why peripheral references narrower than ~ 60–80% are problematic, and why ALICE introduced the "hybrid" ZNA-based centrality estimator for small systems [14].

Practical rule: treat $\sigma_{R_{\mathrm{coll}}}$ as a single *global* (Type C) multiplicative uncertainty on R_{CP}, drawn from the sPHENIX official Glauber tables (typically 8–12% for the most peripheral bin / most central bin combination at sPHENIX, anti-correlated between $\langle N_{\mathrm{coll}}^{\mathrm{cent}} \rangle$ and $\langle N_{\mathrm{coll}}^{\mathrm{peri}} \rangle$ when both move with the trigger efficiency).

3 Catalog of Systematic Uncertainty Sources for sPHENIX π^0 R_{CP}

The following catalog adapts PHENIX precedent [1, 2, 4] and ALICE practice [11, 12, 8] to the specifics of the sPHENIX EMCal (W/SciFi, $|\eta| < 1.1$, $\Delta\eta \times \Delta\phi = 0.024 \times 0.024$ towers grouped into 2×2 blocks) [21, 22]. Each source is given a recommended Barlow-style variation, an expected order-of-magnitude size, and a cancellation behavior in R_{CP}.

3.1 Detector-level

D1. Energy scale. Anchored by the π^0 mass peak via the Tower Slope Method (Wei thesis [5] as referenced on the BNL sPHENIX wiki [6]; the analyzer's `testslope.C` and `Expo_Rebinning.C` macros implement an iterative Gaussian fit followed by an exponential rebinning of the slope distribution). Variation: shift the per-tower scale by the residual after the last calibration iteration (± 0.5–1% for sPHENIX based on the mass-peak stability seen in 2023 commissioning [22]). Propagate by re-running yield extraction with energy scaled $\pm$. *Cancels partially in R_{CP} at the level of the centrality dependence of the residual non-uniformity.*

D2. Energy resolution. Vary the smearing parameters a, b in $\sigma_E/E = a/\sqrt{E} \oplus b$ within their measured range (test-beam [21]: $a \approx 13\%$, $b \approx 2.5\%$ in beam, degraded in situ). Re-derive efficiency and bin-migration corrections from MC for each smearing variant.

D3. Tower-by-tower calibration. The Tower Slope Method has finite per-tower statistics; the residual non-uniformity is the RMS of the per-tower mass-peak position after convergence. Propagate by replaying the analysis with a Gaussian-distributed per-tower scale fluctuation matched to the measured RMS. *Largely cancels in R_{CP}* since both numerator and denominator share the same calibration map; what *does not* cancel is any centrality-dependent occupancy effect on the slope (item D6).

D4. Position resolution. Affects the opening-angle reconstruction. Vary the cluster-position algorithm (logarithmic-weighted vs. linear centroid, $w_0 \in \{4.0, 4.5, 5.0\}$).

D5. Dead and hot tower masking. Vary the mask threshold ($\pm 1\sigma$ on the per-tower mean energy distribution), and check that mass peak position is invariant under the variation. PHENIX assigned 1–2% from this [1].

D6. Occupancy-driven shifts. In central Au+Au, multiple-cluster overlaps in the EMCal can systematically shift the reconstructed mass and width relative to peripheral. Quantify by fitting $\mu_{\pi^0}(c)$ and $\sigma_{\pi^0}(c)$ vs. centrality and assigning the centrality-dependent residual as a Type-B uncertainty that does *not* cancel in R_{CP}.

3.2 Reconstruction-level

R1. Clusterizer choice. Compare the default island/v3 clusterizer to a fixed-3×3 tower window. Take the difference as a systematic.

R2. Cluster-energy threshold $E^{\mathrm{min}}_{\mathrm{clus}}$. Vary $\{0.5, 1.0, 1.5\}$ GeV. The 2023 sPHENIX commissioning analysis [22] used ≥ 1 GeV.

R3. Splitting/merging cuts. For high-p_T π^0 above ~ 10 GeV/c, the two photons begin to merge; the analyzer's framework should evaluate split-cluster recovery with a varied minimum-distance cut.

R4. Photon ID cuts. Shower-shape χ^2 (or σ^2_{long} in PHENIX/ALICE language [12]), cluster timing $|t_{\mathrm{clus}}| < 10$ ns, charged-track veto. Vary each cut by $\pm 1\sigma$ around the nominal.

R5. Asymmetry cut $\alpha = |E_1 - E_2|/(E_1 + E_2)$. The standard sPHENIX/ALICE choice $\alpha < 0.7$–0.8 [11] should be varied ± 0.1.

3.3 Signal extraction

S1. Background functional form. The analyzer's framework already supports Gauss+pol1/pol2/pol3 plus a power-law variant; the standard PHENIX practice is to take the half-spread across the polynomial-order set as the systematic [2].

S2. Fit range. Vary the lower edge $m^{\mathrm{lo}} \in \{0.07, 0.08, 0.09\}$ GeV/c^2 and upper edge $m^{\mathrm{hi}} \in \{0.20, 0.22, 0.25\}$ GeV/c^2; take RMS of the resulting yields.

S3. Mass-window integration. Vary $\pm 2\sigma \to \pm 3\sigma$ around the fitted peak. Equivalent to integration vs. fit-yield comparison.

S4. Sideband definition. For the iterative sideband-based χ^2 that the analyzer uses, vary the inner and outer sideband edges symmetrically ± 1 bin.

S5. Signal shape. Replace the single Gaussian by (i) a Crystal Ball with the tail parameter from MC, and (ii) a sum of two Gaussians (core + tails). The half-spread is the systematic. ALICE 7 TeV pp analysis assigned ~ 1–3% from this [12].

S6. Mixed-event vs. same-event combinatorial subtraction. If both are implemented, the difference is a systematic; this is one of the cleaner cross-checks because mixed-event background carries a different normalization bias than the polynomial sideband.

3.4 Efficiency and acceptance

E1. Single-particle MC vs. embedded MC. The `pi0Efficiency` Fun4All module produces $\varepsilon \times A(p_T)$ from single-π^0 throws. The systematic from underlying-event distortion is bounded by the difference between this and an embedded sample (HIJING + GEANT4 + reconstruction). *This difference is centrality-dependent and is the dominant non-cancelling efficiency systematic for R_{CP}.*

E2. p_T shape weighting. The MC truth p_T spectrum must match the data. Iteratively reweight the MC truth to the unfolded data spectrum and assign the difference of $\varepsilon(p_T)$ between iterations as a systematic. PHENIX assigned 1–3% [1].

E3. Material budget / conversion correction. For sPHENIX, the EMCal sits outside MVTX/INTT/TPC, so a non-trivial fraction of $\pi^0 \to \gamma\gamma$ photons convert before reaching the calorimeter. The X/X_0 inside the TPC outer field cage is presently ~ 8–10%. Vary by $\pm 5\%$ relative.

E4. Trigger / minimum-bias selection. If a high-p_T photon trigger is used to extend reach, the trigger turn-on must be measured from the MB sample. PHENIX practice is to fit an error-function turn-on and assign the shape as a systematic above the plateau.

3.5 Centrality and $N_{\rm coll}$

C1. Glauber MC parameters ($\sigma_{\rm inel}^{NN}$, $r_{\rm WS}$, $a_{\rm WS}$, NBD parameters, ancestor-model variants). Take the standard sPHENIX Glauber table central value and uncertainty band [17, 18].

C2. MBD / sEPD trigger efficiency for pp-like events. Drives the largest contribution at peripheral. Treat as Type C with anti-correlation between $\langle N_{\rm coll}^{\rm cent}\rangle$ and $\langle N_{\rm coll}^{\rm peri}\rangle$.

C3. Centrality definition. Vary the centrality estimator: charge sum in MBD vs. sEPD vs. ZDC-based hybrid, following ALICE precedent [14]. In Au+Au this is a small but nonzero check.

C4. Peripheral interval choice. Compare $[60, 80]\%$ and $[60, 92]\%$ (PHENIX historical) and $[50, 80]\%$. For each peripheral choice, the peripheral-bin geometric bias [13] should be quoted explicitly: in $[60, 92]\%$ Pb+Pb at LHC, ALICE found that $R_{AA}^{\rm peri}$ deviates from unity by $\sim 25\%$ *purely from selection bias and an incoherent-superposition assumption that fails for very peripheral collisions.*

3.6 Pile-up, run-by-run stability, miscellaneous

O1. In-time pile-up / event multiplicity bias. Compare yields in low-luminosity vs. high-luminosity run subsets.

O2. Out-of-time pile-up. sPHENIX EMCal SiPM signals integrate over a finite time window; quantify by varying the timing cut.

O3. Run-by-run stability. Compute the mean yield per event in fine run groups; assign the RMS / $\sqrt{N_{\rm run}}$ as a Type-B systematic.

O4. η feed-down to π^0. Subtract the secondary π^0 contribution from K_S^0 and η decays using HIJING + GEANT4. Assign the model-dependent uncertainty.

4 Statistical Analysis Plan

Parker (PI judgment)

Tasks Parker owns. Choosing which fit functional form is "default", choosing the peripheral interval (with documented physics rationale), defining what counts as a passed closure test, and signing off on the final correlation pattern of the systematics.

4.1 Yield extraction uncertainty from fit covariance

For each (p_T, c) bin, perform an extended unbinned likelihood fit (or, equivalently for the high-statistics regime, a binned χ^2). The yield $\hat{N}_S$ and its covariance with the background nuisance parameters $\boldsymbol{c}$ should be retained from the `TFitResult` object in ROOT and serialized for downstream use:

$$V_{\hat{N}_S} = \left(\frac{\partial N_S}{\partial \theta_k}\right) V_{kl}^{\theta} \left(\frac{\partial N_S}{\partial \theta_l}\right). \tag{7}$$

4.2 Bootstrap and toy MC procedures

Two complementary procedures are recommended:

1. **Nonparametric bootstrap**: resample with replacement the unbinned $m_{\gamma\gamma}$ candidate list per (p_T, c) bin, refit, take the RMS of the yield distribution. This captures both Poisson statistics and fit instability.

2. **Parametric toy MC**: sample $\boldsymbol{\theta}$ from the fit covariance, regenerate $m_{\gamma\gamma}$ histograms, refit. This tests pull behavior and checks for fit bias.

A minimum of 10^3 toys per bin should be run; the analyzer's `BM.C` plotting infrastructure can be extended to ingest toy ensembles. This is a natural agentic-assist workload (Sec. 6.3).

4.3 Likelihood-based combination across p_T bins

When comparing the measured $R_{\rm CP}(p_T)$ to a theoretical model curve, use the full covariance:

$$\chi^2_{\rm model}(\boldsymbol{\lambda}) = \left(\boldsymbol{R}_{\rm CP}^{\rm data} - \boldsymbol{R}_{\rm CP}^{\rm model}(\boldsymbol{\lambda})\right)^T \left({\rm Cov}^{\rm tot}\right)^{-1} \left(\boldsymbol{R}_{\rm CP}^{\rm data} - \boldsymbol{R}_{\rm CP}^{\rm model}(\boldsymbol{\lambda})\right), \tag{8}$$

where $\boldsymbol{\lambda}$ are the model parameters (e.g., $\hat{q}$). Marginalize the global Type-C systematic over a Gaussian nuisance parameter; profile the likelihood for $\boldsymbol{\lambda}$.

4.4 Low-statistics peripheral bins

Validation & Warning

At $p_T \gtrsim 15$ GeV/c in $[60, 80]$%, the expected π^0 count per bin will be $\mathcal{O}(10)$. A χ^2 fit fails Gaussian asymptotics; switch to a profile-likelihood (Feldman–Cousins-style) or a Poisson-extended binned ML. Confirm yield-extraction stability against (i) rebinning of p_T, (ii) widening to $[50, 80]$% as a higher-statistics cross-check. *Do not* smooth the peripheral spectrum before forming $R_{\rm CP}$; this introduces a pull bias that does not commute with model comparison.

5 Validation and Closure Tests

1. **MC closure of $\varepsilon \times A(p_T)$.** Throw a flat-in-$p_T$ single-π^0 MC, reconstruct, divide reconstructed-by-truth, verify that applying this efficiency to a separate test sample recovers the truth p_T spectrum within statistics.

2. **Single-particle vs. embedded MC comparison.** The fractional difference $\Delta\varepsilon^c(p_T) \equiv [\varepsilon^c_{\rm emb} - \varepsilon^c_{\rm single}]/\varepsilon^c_{\rm single}$ should be benign in peripheral, growing in central; the centrality dependence *is* the systematic.

3. **Mass peak diagnostics vs. centrality.** Fit $\mu_{\pi^0}(c)$ and $\sigma_{\pi^0}(c)$. PHENIX showed μ_{π^0} stability at the $\lesssim 1$% level in 0–92% [2]; sPHENIX should target the same.

4. **Integrated yield consistency.** Sum $\int dp_T\, dY/dp_T$ over all centralities; compare against a Glauber-scaled pp expectation. A spectacular failure is the canary.

5. **Cross-method comparison** (where applicable). If a PCM-style track-converted-photon analysis becomes available in sPHENIX (after MVTX/INTT integration), cross-checking is gold-standard, as ALICE has demonstrated [11, 8, 7].

6. **Benchmarking against PHENIX.** The 200 GeV Au+Au π^0 R_{AA} shape from PRC 87 034911 [1] is the gold standard for sPHENIX to reproduce within uncertainties when the per-event yields are formed against the appropriate *pp* baseline. Likewise, the central-to-peripheral ratio of yields should be compatible.

7. η/π^0 **ratio.** Should be $\sim 0.46 \pm 0.06$ at high p_T [1], centrality- and $\sqrt{s_{NN}}$- independent; a powerful internal cross-check that does not require cross-comparison to other experiments.

6 Workflow Design: Parker vs. Agents

The recent literature on LLM-driven HEP analysis [28, 29, 30, 31] converges on three principles that this plan adopts:

1. The agent operates inside a deterministic workflow manager (Snakemake, Make, or a Fun4All driver script). The agent generates and corrects code; the workflow manager guarantees reproducibility.

2. Every agent action is logged, version-controlled (git), and produces reproducible artifacts (input hashes, output hashes).

3. Agent failure rates remain non-trivial ($\sim$ 40–50% on data-prep tasks in Gendreau-Distler *et al.* [28]). Treat agent output as draft code requiring review, not as authoritative.

6.1 Parker-owned tasks

Parker (PI judgment)

Non-delegable.

- Defining the analysis “network”: which centrality classes, which p_T binning, which peripheral reference, which baseline systematic categorization.
- Physics judgment on cut definitions, rationale, and trade-offs.
- Final assignment of systematic uncertainty values, including the choice of Barlow-criterion threshold for retaining a variation.
- Interaction with the sPHENIX paper preparation group (PPG) and STAR PWG: presenting plots, defending choices, responding to peer questions.
- Thesis writing, including narrative motivation, physics interpretation, and connection to theoretical frameworks ($\hat{q}$ extraction, BDMPS, GLV, etc.).
- Collaboration-level reproducibility: ensuring the analysis can be re-run by a successor student in 5 years.
- Critical reading of agent output: if the agent generates a fit macro that subtly mishandles bin centering, only physics judgment catches it.

6.2 Agent-strong tasks

Agent-assisted

Where Claude / LLM agents add real leverage.

- **ROOT/C++ macro generation and refactoring**: e.g., expanding `testslope.C` to scan additional rebinning schemes, generating a templated systematic-variation driver from a single nominal macro.
- **Batch systematic-variation runs**: looping the same fit over {pol1, pol2, pol3, power-law} $\times$ {3 fit ranges} $\times$ {3 mass windows} = 36 variants per (p_T, c) bin. Agent generates a manifest of jobs, dispatches, monitors, and aggregates RMS/half-spread numbers into a single LaTeX table.
- **Regression checks across versions**: when the underlying Fun4All version updates, agent compares yield extraction outputs bin-by-bin and flags any deviation > statistical noise.
- **Literature search and citation gathering**: scanning INSPIRE-HEP and arXiv for related theses and notes; producing annotated BibTeX.
- **Drafting analysis notes / sections of the thesis**: generating first-draft LaTeX from notes, building TikZ flowcharts, populating tables.
- **Debugging**: ROOT-specific traceback interpretation (memory leaks, `TFile` ownership), Fun4All module wiring.
- **Cross-check macro generation**: e.g., automatically translating a PHENIX-style η/π^0 check macro to sPHENIX I/O conventions.
- **Uncertainty table assembly**: ingesting the per-source variation outputs, applying the type A/B/C labels, producing the final correlated-vs-uncorrelated breakdown table in the thesis format.

6.3 Concrete agent-assisted sub-pipelines

Figure 1 sketches the integration. Each numbered node is either a *decision* (Parker-owned, blue) or a *mechanical execution* (agent-eligible, teal).

Sub-pipeline (a): Batch systematic variation. A YAML/JSON manifest lists each variation with its label and parameter overrides. The agent generates a templated ROOT macro `run_variation_$LABEL.C` for each row, submits to the sPHENIX RCF batch system, and ingests output histograms into a SQLite or pandas DataFrame keyed by $(p_T, c, \text{variation})$. Parker reviews the manifest *before* dispatch. Output: a single canonical "systematics aggregate" file.

Sub-pipeline (b): Automated fit-quality monitoring. The agent flags fits with $\chi^2/\mathrm{NDF} > 2$, fit-converged status `kFALSE`, or yield error fraction > 50%. These get a per-bin diagnostic plot auto-attached to a daily Slack/email report. Parker manually inspects flagged bins.

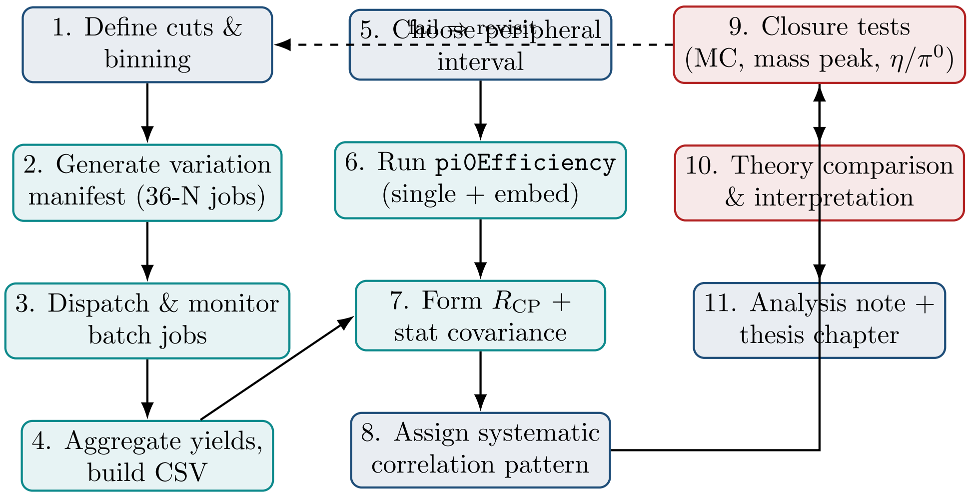

Figure 1: Analysis pipeline with explicit agent integration points. Blue = Parker-owned decision; teal = agent-eligible mechanical execution; red = validation gate.

Sub-pipeline (c): Cross-check macro generation. The agent translates an analysis-note prescription (in plain English) into an executable ROOT macro shell, with the physics-relevant lines as TODOs for Parker to fill in. The agent does *not* silently invent cuts.

Sub-pipeline (d): Uncertainty table assembly. Given the aggregate file from (a) and a YAML mapping of source→type, the agent emits a LaTeX longtable matching the thesis style (Sec. 7). This is mechanical and a natural agentic task; the validation is that Parker reviews the YAML mapping (the physics judgment) and then accepts the auto-generated table.

6.4 Validation discipline

Validation & Warning

Hard rules for agent use in this analysis.

1. Every agent-generated macro is reviewed line-by-line on first generation and committed to git with a commit message that includes the prompt.
2. Agent code is run in a containerized, deterministic environment (a Singularity image with the pinned sPHENIX build). A given prompt + container ⇒ a given output, modulo LLM stochasticity which is logged.
3. Closure tests (Sec. 5) are gating: an analysis cannot "pass" on agent-generated code that has not been independently re-derived by Parker on at least one nominal bin.
4. Numbers in the thesis come from a frozen, signed-off pipeline run. The agent is allowed to draft text but never to insert numerical values without a traceable provenance chain.
5. Any agent output that disagrees with published PHENIX/ALICE precedent at the factor-of-two level is treated as a *bug* until proven otherwise.
6. LLM hallucination of references is a known failure mode [31, 28]; *every* citation must be cross-checked against arXiv/INSPIRE-HEP by Parker before inclusion.

7 Thesis-Style Uncertainty Budget Template

The placeholder values below are scaled from PHENIX 200 GeV Au+Au precedent [1, 2]; *actual values must be derived from sPHENIX data and replace these.* Centrality columns are 0–10% (cent.) and 60–80% (peri.) and the R_{CP} column accounts for cancellations.

Table 1: Template uncertainty budget for sPHENIX π^0 R_{CP} at representative $p_T \approx 8$ GeV/c. Type A is point-to-point uncorrelated; B is correlated within p_T; C is global. Values are placeholders for illustration.

Source (label)	**Type**	**Cent.** [%]	**Peri.** [%]	**Cancel.**	R_{CP} [%]
Detector					
Energy scale (D1)	B	4.0	4.0	partial	1.5
Energy resolution (D2)	B	2.0	2.0	partial	1.0
Tower-by-tower calib (D3)	B	2.5	2.5	yes	0.7
Position resolution (D4)	A	0.8	1.2	no	1.4
Dead/hot tower mask (D5)	B	1.5	1.5	yes	0.4
Occupancy shift (D6)	B	2.0	0.5	no	2.1
Reconstruction					
Clusterizer (R1)	B	2.0	2.0	yes	0.6
$E_{\mathrm{clus}}^{\mathrm{min}}$ (R2)	B	1.0	1.0	yes	0.3
Splitting/merging (R3)	B	1.5	1.0	partial	1.0
Photon ID cuts (R4)	B	2.5	2.0	partial	1.5
Asymmetry α (R5)	B	1.5	1.5	yes	0.4
Signal extraction					
Background pol order (S1)	A	3.0	4.0	no	5.0
Fit range (S2)	A	1.5	2.5	no	2.9
Mass-window integration (S3)	A	1.0	1.5	no	1.8
Sideband definition (S4)	A	1.0	2.0	no	2.2

Table 1: Template uncertainty budget for sPHENIX π^0 R_{CP} at representative $p_T \approx 8$ GeV/c. Type A is point-to-point uncorrelated; B is correlated within p_T; C is global. Values are placeholders for illustration.

Source (label)	**Type**	**Cent.** [%]	**Peri.** [%]	**Cancel.**	R_{CP} [%]
Signal shape (S5)	B	1.5	2.0	partial	1.3
Efficiency / acceptance					
Single vs. embedded MC (E1)	B	4.0	1.5	no	4.2
p_T shape weighting (E2)	B	1.5	1.5	yes	0.4
Material / conv (E3)	B	2.0	2.0	yes	0.5
Trigger turn-on (E4)	B	1.0	1.5	partial	0.8
Centrality / N_{coll}					
R_{coll} (C1)	C	–	–	no	8.5
Trigger eff. in *pp*-like (C2)	C	–	–	no	(in C1)
Centrality estimator (C3)	B	1.5	3.0	no	3.4
Peripheral interval (C4)	C	–	–	no	4.0
Other					
Pile-up (O1)	B	1.0	0.5	no	1.1
Out-of-time pile-up (O2)	B	0.5	0.5	yes	0.2
Run-by-run (O3)	B	0.5	1.0	no	1.1
η feed-down (O4)	B	1.0	1.0	yes	0.3
Total uncorrelated (Type A)					**6.7**
Total correlated within p_T (B)					**6.3**
Total global (Type C)					**9.4**
Total in quadrature					**13.4**

Reading the table. The Type-A column is what should be drawn as bin-to-bin error bars on R_{CP} plots; the Type-B and Type-C totals are typically drawn as separate boxes (B-band around the points, C-box at the right edge of the plot, in the PHENIX/ALICE convention).

8 Annotated Bibliography of Recommended PhD Theses, Primary Papers, and Methodology References

The following annotated list is the working bibliography for the analysis note and thesis. Each entry includes a 2–3 sentence note on what it is most useful for in the present analysis context.

PhD theses — PHENIX lineage.

- **R. Wei**, π^0 *measurements with the PHENIX EMCal* (Stony Brook University) [5]. Cited on the BNL sPHENIX wiki [6] as the canonical reference for the Tower Slope Method that the analyzer uses. Read for the rebinning treatment, slope-extraction systematics, and the iterative convergence criteria.

- **B. Sahlmüller**, π^0 *and* η *in* $\sqrt{s_{NN}} = 200$ *GeV Au+Au with PHENIX* (Universität Münster) [4]. The most directly comparable PHENIX EMCal-based π^0 analysis to what sPHENIX is now reproducing. Use as a template for systematic uncertainty categorization, mass-peak diagnostics vs. centrality, and the iterative background-subtraction defense.

- **F. Bock**, *Neutral pion and direct photon measurements with ALICE via PCM and EMCal* (Universität Heidelberg) [7]. The gold-standard ALICE-side reference for combining PCM and calorimeter methods and for the careful treatment of material-budget uncertainty. The systematics tables are pedagogically excellent.
- **L. Leardini**, ALICE Heidelberg, R_{AA} in Pb-Pb at 2.76 TeV [8]. PCM + PHOS combined analysis; treats the polynomial-order variation, mixed-event normalization, and the centrality-dependent α cut explicitly.
- **C. Klein-Bösing** group (Münster) and the broader ALICE-Frankfurt π^0 thesis lineage (recent J. König, N. Schmidt theses) [9, 10]. Relevant for the merged-cluster regime ($p_T \gtrsim 16$ GeV/c) and the high-p_T extension techniques sPHENIX will need.
- **Recent / in-progress sPHENIX theses** (BNL, Yale, Iowa State, UC Riverside, Georgia State, UCLA, etc.) on detector commissioning and first $\pi^0\,v_2$ [22]. Track the BNL sPHENIX wiki and the sPHENIX speakers' bureau page for emerging thesis work.

Primary collaboration papers.

- PHENIX, **Phys. Rev. C 87, 034911** (2013) [1]. The fourfold-luminosity Au+Au π^0 paper that sets the high-p_T benchmark. Provides per-centrality R_{AA}, the energy-loss fractional-loss extraction, and the v_2-vs-p_T azimuthal anisotropy. The systematic uncertainty tables (their Table II) are the closest analog to what should appear in the thesis.
- PHENIX, **Phys. Rev. Lett. 101, 232301** (2008) and the follow-up **nucl-ex/0611007** [2]. The detailed systematic-uncertainty methodology for PHENIX EMCal π^0 measurements; many constructs (signal-region/sideband mass-window dichotomy, conversion correction, type A/B categorization) are inherited.
- PHENIX, **Phys. Rev. C 88**, evolution of π^0 suppression [3]. Useful for the $\sqrt{s_{NN}}$-dependence and for the comparison of R_{CP} between RHIC energies, which sPHENIX can extend.
- ALICE, **Eur. Phys. J. C 74, 3108** (2014) [11]. Pb-Pb 2.76 TeV π^0 R_{AA} via PCM and PHOS. Crystallizes the PCM-vs-calorimeter cross-method comparison.
- ALICE, **Eur. Phys. J. C 78, 624** (2018), π^0 and η in p-Pb [12]. The systematic uncertainty section is particularly rich for the calorimeter-photon-ID and the secondary-π^0 correction; recommended bedside reading.
- ATLAS, π^0/jet R_{CP} in p-Pb [15, 16]. Definitive recent treatment of R_{CP} at LHC, including the careful definition of R_{coll} and centrality-estimator dependence.
- ALICE peripheral-bias analysis [13]. Mandatory reading on the geometric and selection biases of $R_{AA}^{\mathrm{peri}} \neq 1$.

sPHENIX collaboration documents.

- sPHENIX TDR [19], especially §7.1 on EMCal physics and calibration — which the analyzer is already reading.
- sPHENIX Beam Use Proposal 2024 [20]. Quantifies expected statistics for Run-25 Au+Au and the $p+p$ baseline.

- EMCal prototype beam-test paper [21]. Source of the canonical $\sigma_E/E = 16\%/\sqrt{E} \oplus 5\%$ resolution.
- sPHENIX first-results paper, Nouicer (2024) [22], with $\pi^0\, v_2$ in Run-23 commissioning. This is the methodological precursor that establishes the sPHENIX cluster ID and signal-extraction defaults.
- Tannenbaum's analysis [17]. Critical reading on the binomial trigger-efficiency reduction in central A+A and how it propagates to $N_{\mathrm{part}}/N_{\mathrm{coll}}$ uncertainty.
- PHENIX Glauber MC note [18], BNL internal. Template for what the sPHENIX official Glauber table should contain.

Statistical methodology.

- R. Barlow, "Systematic errors: facts and fictions" (hep-ex/0207026) [23]. The taxonomy and the famous "Thou shalt not" commandments. Read every two years.
- P. Sinervo, "Definition and treatment of systematic uncertainties in HEP" [25]. Practical guide to the type A/B/C categorization that this plan adopts.
- G. Cowan, *Statistical Data Analysis* (Oxford, 1998) [26]. Reference for nuisance-parameter likelihoods, profile likelihood, asymptotic formulae; chapters 6 and 9 in particular.
- R. Barlow, "Asymmetric systematic errors" (hep-ph/0306138) [24]. For when systematic variations are visibly asymmetric.
- PDG Review of Statistics (2024 edition) [27]. The current authoritative summary; cite this in the thesis.

LLM / agentic workflows in HEP.

- Gendreau-Distler et al., "Automating HEP Data Analysis with LLM-Powered Agents" (arXiv:2512.07785, NeurIPS ML4PS 2025) [28]. The LLM+Snakemake architecture this plan adapts; failure-mode taxonomy (data-prep, ROOT-file-context errors) is directly relevant. Their $\sim 57\%$ end-to-end completion rate sets the realistic expectation.
- Esmail et al. / Moreno et al. / LangGraph-based document-grounded analysis [29]. Useful template for an agent that ingests an analysis note (the user's BM.C analysis note) and produces a code shell.
- IRIS-HEP *LLMs as Assistants* project [30]. Umbrella effort with current best-practice guidelines for HEP-specific LLM deployment; subscribe to their blueprints.
- Rasheed et al., "AI-powered code review with LLMs" [31]. Treats the code-review use case specifically; documents both strengths and known failure modes (hallucinated APIs, missed semantic bugs).

9 Suggested 12-Month Timeline

The timeline below assumes a starting point of *April 2026*. Months are nominal; actual cadence depends on the sPHENIX Run-25 calibration freeze. "A:" marks a primary agentic-assistance opportunity.

Month	Parker-led milestone	Agent-assisted task
M1 (Apr)	Finalize centrality binning, peripheral choice, p_T binning. Lock cut definitions. Write nominal-version analysis note skeleton.	A: Generate templated systematic-variation manifest (YAML); refactor `testslope.C` and `Expo_Rebinning.C` for batch use; build CI tests for nominal yield extraction.
M2 (May)	Tower Slope Method: produce final calibration map; declare the residual non-uniformity. Approve the closure target.	A: Sweep tower-by-tower fit ranges; auto-flag towers with $\chi^2/\mathrm{NDF} > 2$; produce per-sector calibration QA pages.
M3 (Jun)	Run `pi0Efficiency` on single-particle MC across all centralities; commit nominal $\varepsilon \times A(p_T, c)$. Validate by closure on independent MC.	A: Auto-build embedded-MC vs. single-particle comparison plots; emit per-bin difference table.
M4 (Jul)	Implement Glauber + N_{coll} table for sPHENIX MBD-based centrality; consult sPHENIX centrality WG.	A: Bootstrap N_{coll} uncertainty over Glauber parameter set (10^4 throws); produce official table.
M5 (Aug)	Run nominal yield extraction over data; first-look R_{CP} plot (statistical errors only). Internal review with advisor.	A: Job dispatcher for batch-extraction over runs; aggregate to single ROOT file.
M6 (Sep)	Define Barlow-criterion threshold for keeping each systematic variation. Meet with sPHENIX hard-probes WG to pre-register the systematic plan.	A: Run all ~ 36 signal-extraction variations; produce S1–S5 systematic per-bin tables.
M7 (Oct)	Detector-level systematics (D1–D6). Re-run with $\pm$ energy scale; assess occupancy shift.	A: Generate $\pm$-scale histograms; build TikZ figure of mass-peak vs. centrality and vs. run group.
M8 (Nov)	Reconstruction-level systematics (R1–R5); cross-check with embedded MC.	A: Cross-method comparison plots; auto-format for analysis note.
M9 (Dec)	Centrality / N_{coll} systematics (C1–C4); peripheral-interval cross-check ($[60, 80]$ vs $[60, 92]$ vs $[50, 80]$).	A: Build covariance matrix from per-source per-bin variations; produce $\mathrm{Cov}_{ij}^{\mathrm{tot}}$ heatmap.
M10 (Jan)	Closure tests (Sec. 5). Mass-peak vs. centrality, η/π^0 ratio. Internal review II.	A: Auto-generate closure-test plots and a "red-light/green-light" summary page.
M11 (Feb)	Theory comparison: $\hat{q}$ extraction or shape comparison to JEWEL/SCET-G/MARTINI. Begin paper proposal in PPG.	A: Draft TeX of analysis note from notes; produce PPG-ready plots; literature search for newest theory predictions to cite.
M12 (Mar)	Thesis chapter draft; PPG presentations.	A: LaTeX table assembly (Sec. 7); citation cross-check; figure-caption drafts.

Parker (PI judgment)

The timeline puts *three internal-review gates* at M5, M10, and M12. Each gate is a Parker-and-advisor sign-off on a frozen pipeline output. *No agent-generated artifact crosses a gate without explicit Parker review.*

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Compiled April 29, 2026 for P. Lewis, Ohio University. This document is intended as an internal working framework; numbers in Sec. 7 are placeholders. All systematic-uncertainty values must be derived from the actual sPHENIX dataset and pipeline before being reported externally.